Stochastic Processes

The definition, unpacked ingredient by ingredient

Abstract

A stochastic process is a random function. One random draw selects an entire path of values over all times at once, not a single number. This chapter builds the formal definition from that idea and shows which parts of it a practitioner uses.

The definition has four ingredients: a probability space as the source of randomness, an index set of times or locations, a state space for the values, and a measurability condition. Properties such as the Markov property are extra assumptions added to this definition. A process has two readings. Fixing a time gives a random variable: the distribution of values at that moment. Fixing an outcome gives a sample path: an ordinary function with no randomness left. A simulation builds a path step by step, but the definition fixes the whole path with one draw. The values at all times come from the same draw. This shared draw gives joint events, such as a low value today and a high value next month, their probabilities. Without it, the theory could not relate one moment to another.

Probability assigns sizes to sets of outcomes, not weights to single outcomes. For a number drawn uniformly between 0 and 1, every single value has probability zero, yet some value always occurs. Measure theory handles this case. Length, area, mass and probability are one idea. Probability is the case where the whole space has size one. No consistent rule can size every subset of the unit interval, so the theory fixes in advance the collection of sets it will size. The measurability condition requires every question about the values, such as whether the price is below 90, to correspond to one of those sets. If the condition fails, the probability is undefined. It is neither zero nor unknown.

In practice, a model specifies a process by its finite-dimensional distributions: the joint behavior of the values at any finite set of times. If these distributions are consistent, the Kolmogorov extension theorem guarantees that a process with that behavior exists. A model specifies Brownian motion this way. One-time marginal distributions are not enough. Finite-dimensional distributions do not decide whether paths are continuous. A filtration records what is known at each time, and it only grows. A process is adapted if its value at each time is known at that time. Adaptedness is the formal version of the backtesting rule against lookahead bias.

How to read this document. Part I (Sections 1–9) is the main text. Sections 1–5 explain the formal definition of a stochastic process. The definition has four ingredients, and each one gets a section. Sections 6–8 cover how the definition is used: the two ways to read a process, the finite-dimensional distributions that identify it, and the filtration that almost every application adds. Section 9 applies the definition to five standard processes and collects the equivalences between its objects. Part II, the appendix (Sections A–K), defines the measure-theoretic terms from the beginning. It covers σ-algebras, measurable spaces, measures, probability spaces, random variables, integration, conditioning, and product spaces. A reader who knows measure theory can skip Part II. A reader who does not can read it first, or consult it whenever Part I uses an unfamiliar term.

Objectives. After this chapter, you should be able to:

- state the definition of a stochastic process and name the job of each ingredient.
- read a process two ways, as a random variable at each time and as a random function of time.
- explain why measurability decides whether a probability exists at all.
- say what the finite-dimensional distributions determine about a process, and what they leave open.
- express “no lookahead” as adaptedness to a filtration, and check a backtest signal against it.

Epistemic tags. The chapters in this collection flag claims by status:

- **[Fact]** — replicated across independent datasets or implementations, with broad agreement.
- **[Contested]** — documented, but with ongoing disagreement about magnitude, robustness, or cause.
- **[Hypothesis]** — a proposed mechanism, not decisively tested.
- **[Practice]** — practitioner convention. The evidence may be private or absent. A [Practice] claim has not been debunked.

This chapter is mathematics. Its definitions and theorems are facts by construction, so they carry no tag. The one tag that appears is [Practice], on conventions that textbooks assume rather than prove.

Notation. The table lists every symbol that recurs in the chapter, with the section that defines it. Symbols used in a single example are defined where they appear.

Symbol	Meaning	Defined in
$(\Omega, \mathcal{F}, \mathbb{P})$	Probability space: sample space, σ-algebra of events, probability measure	§2
ω	An outcome, one element of Ω: a complete realization of everything random	§2
T; s, t; n	Index set; indices in it (times, with $s \leq t$ when ordered); an index in discrete time, or the number of times $t_1, \dots, t_n$ in §7	§3
$(E, \mathcal{E})$	State space: the set of possible values, with its σ-algebra	§4
$X = (X_t)_{t \in T}$	The stochastic process; $X_t : \Omega \to E$ is its value at index t	§1
A, B, A_n	Measurable sets; in Part I, B is a set of values: a member of $\mathcal{E}$, or in §7 a measurable subset of E^n	§5, B
$X_t^{-1}(B)$	Preimage: the outcomes ω with $X_t(\omega) \in B$	§5
E^T	Path space: all functions from T to E, with the product σ-algebra	§7, K
$\mu_{t_1 \dots t_n}$	Finite-dimensional distribution: the joint law of $(X_{t_1}, \dots, X_{t_n})$	§7
$\mathcal{N}(m, v)$	Normal distribution with mean m and variance v	§7
W_t; N_t	Brownian motion; a counting process, such as the Poisson process	§7, §4
$(\mathcal{F}_t)_{t \in T}$; $\mathcal{F}_t^X$	Filtration; natural filtration of X	§8
$(U, \mathcal{U}, \mu)$, $(V, \mathcal{V}, \nu)$	Generic measurable spaces with measures, used in Part II	B, E
2^U	Power set: every subset of U	B
$\sigma(\mathcal{C})$; $\sigma(X)$	σ-algebra generated by a collection of sets $\mathcal{C}$; by a random variable X, or by a family such as $(X_s)_{s \leq t}$	C
$\mathcal{B}(\mathbb{R})$	Borel σ-algebra: generated by the open subsets of $\mathbb{R}$	C
λ; δ_x	Lebesgue measure (length); Dirac measure at the point x	E
a.s.	Almost surely: outside a set of probability zero	F
f, g	Generic measurable functions; in H, f is a density	G
μ_X; F_X	Law of X; its cumulative distribution function	H
$\mu \ll \nu$	μ is absolutely continuous with respect to ν	H
$\mathbb{1}_A$; $\mathbb{1}\{\cdot\}$	Indicator of A: equal to 1 on A and 0 elsewhere; indicator of the condition in braces	§7, I
$\mathbb{E}[X]$; $\mathbb{E}[X \mid \mathcal{G}]$	Expectation; conditional expectation given a σ-algebra $\mathcal{G}$	I, J
$\mathbb{N}$; $\mathbb{N}_0$; $\mathbb{Z}$	$\{1, 2, \dots\}$; $\{0, 1, 2, \dots\}$; the integers	—

Two symbols need a warning. In this chapter, σ appears only in the names "σ-algebra" and "σ-finite" and in the operator $\sigma(\cdot)$. It never denotes volatility, which this chapter does not use. And $\mathbb{E}$, in blackboard bold, is expectation, while the italic E is the state space.

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Part I — The definition

1. The definition

A stochastic process is a collection of random quantities, one for each index, all driven by the same source of randomness. The formal definition makes each part of that sentence precise.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $(E, \mathcal{E})$ be a measurable space, and let T be a non-empty set. A **stochastic process** with index set T and state space $(E, \mathcal{E})$ is a family of maps

$$X = (X_t)_{t \in T}, \qquad X_t : \Omega \longrightarrow E,$$

such that each X_t is $\mathcal{F}/\mathcal{E}$-measurable:

$$X_t^{-1}(B) \in \mathcal{F} \quad \text{for every } B \in \mathcal{E} \text{ and every } t \in T.$$

In words, each X_t turns an outcome ω into a value in E. The measurability condition says that for every set of values B the theory admits, the outcomes that land in B form an event, so they have a probability.

An equivalent form is often more convenient. A stochastic process is a single map $X : T \times \Omega \to E$ such that $X(t, \cdot)$ is measurable for each fixed t. Fixing the first argument at t recovers the map X_t above.

The definition has four ingredients: three spaces and one condition on the maps between them. Sections 2–5 take them in order.

Ingredient	Symbol	Job	Section
Probability space	$(\Omega, \mathcal{F}, \mathbb{P})$	Holds the randomness	§2
Index set	T	Labels the members of the family, usually by time	§3
State space	$(E, \mathcal{E})$	Holds the values, and fixes which sets of values can be asked about	§4
Measurability	$X_t^{-1}(B) \in \mathcal{F}$	Makes every probability about X_t defined	§5

Three ideas recur in the rest of the chapter. Every section returns to them:

1. **One draw fixes the whole path.** A single outcome ω determines $X_t(\omega)$ for every t at once. A process is a random function, not a sequence of separate random draws.
2. **Every X_t is defined on the same Ω.** Because all the X_t share one probability space, a joint event across times, such as "low today and high next month," has a probability. Dependence across time is defined through it.
3. **A σ-algebra is a list of answerable questions, and also a body of information.** Measurability makes $\mathbb{P}(X_t \in B)$ defined. A filtration, a growing family of σ-algebras, records what is known at each time.

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2. Ingredient I: the probability space

$(\Omega, \mathcal{F}, \mathbb{P})$: **where the randomness is defined, drawn once**

The probability space has three parts, and each has its own job.

Object	Name	What it does
Ω	Sample space	The set of all possible outcomes. One outcome $\omega \in \Omega$ describes how everything turned out: the entire history, not the value at one moment.
$\mathcal{F}$	σ-algebra	The subsets of Ω that count as events, meaning the subsets that receive a probability. It contains Ω and is closed under complement and countable union.
$\mathbb{P}$	Probability measure	Assigns each event a number in $[0, 1]$, with $\mathbb{P}(\Omega) = 1$. The probabilities of countably many disjoint events add up.

$\mathcal{F}$ usually cannot contain every subset of Ω. The uniform probability on $[0, 1]$ shows why. It gives a set the same probability after the set shifts along $[0, 1]$. No countably additive extension of it to every subset of $[0, 1]$ keeps that property (Section A). So $\mathcal{F}$ is deliberately smaller than the set of all subsets. $\mathbb{P}$ answers questions only about members of $\mathcal{F}$.

> **Every X_t is defined on the same $(\Omega, \mathcal{F}, \mathbb{P})$.** The shared space distinguishes a stochastic process from a collection of random variables defined on separate spaces.

The values at all times share one space, so a joint event such as $\{X_1 \leq 3\} \cap \{X_5 > 7\}$ is a subset of the same Ω. Each of the two sets is an event, because X_1 and X_5 are measurable (§5), and $\mathcal{F}$ is closed under intersection. So the joint event belongs to $\mathcal{F}$ and has a probability. Dependence across time is a property of events in this one space. The correlation between X_1 and X_5, or the probability that a price falls and then recovers, describes how events in a single Ω overlap.

Example: coin flips. Let $\Omega = \{0, 1\}^{\mathbb{N}}$ be the set of all infinite sequences of coin flips, with 1 for heads and 0 for tails. Let $\mathbb{P}$ make the flips fair and independent. One outcome ω is an entire infinite sequence. Define $X_n(\omega)$ as the number of heads among the first n flips of ω. No further draw occurs. X_{10} and X_{11} are correlated because both are functions of the same ω and share its first ten flips. Their correlation is $\sqrt{10/11} \approx 0.95$.

§2 Key takeaways

1. One outcome ω is one complete history. Drawing ω once fixes the value of the process at every time.
2. All the X_t are defined on the same probability space. Joint events across times are therefore events with probabilities, and dependence across time can be measured.
3. The σ-algebra $\mathcal{F}$ lists the subsets of Ω that receive a probability. In continuous models it is smaller than the set of all subsets, because some subsets cannot be given a consistent probability.

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3. Ingredient II: the index set T

Usually time, but the definition never says so

T is any set that labels the members of the family. The axioms say nothing about order, continuity, or time. Applications add those properties because the index usually represents time.

The table lists the common choices.

T	Regime	Examples
$\{0, 1, 2, \dots\}$	Discrete time	Daily closing prices, the steps of a Markov chain, the terms of a time series
$[0, \infty)$	Continuous time	Brownian motion, the Poisson process, a diffusion
$\mathbb{R}^2$ or $\mathbb{Z}^2$	Random field	Ore grade across a deposit, pixel noise, temperature over a map. The index is a location, not a time
$\{1, \dots, d\}$	Finite	A random vector with d components. A random vector *is* a stochastic process, and the general definition only widens T.

Most of the technical difficulty in the subject grows with the size of T. With T countable, the definition causes no trouble. A statement about the whole path, such as "the path stays below 5," is a countable intersection of statements about single times. σ-algebras are closed under countable operations. With T uncountable, a statement such as "the path is continuous" constrains uncountably many values at once. Such a statement need not define an event, and continuous-time theory needs extra methods to handle it (§7, Section K).

§3 Key takeaways

1. The index set T can be any set. Time, order, and continuity come from the application, not from the definition.
2. A random vector is a process with a finite index set, and a random field is a process indexed by location.
3. A countable T keeps every statement about the path inside the σ-algebra. An uncountable T does not. That gap is the source of most of the extra methods continuous-time theory needs.

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4. Ingredient III: the state space

$(E, \mathcal{E})$: where the values land, and which sets of values can be asked about

The state space holds the values of the process. It must be a *measurable* space, not just a set. The σ-algebra $\mathcal{E}$ lists the sets of values B for which "is X_t in B?" is a question probability can answer.

The table lists the common choices.

$(E, \mathcal{E})$	Use
$(\mathbb{R}, \mathcal{B}(\mathbb{R}))$	The default: a real-valued process with the Borel σ-algebra, such as a price, a temperature, or a residual.
$(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$	Vector-valued: a particle's position, or the yields at d fixed maturities observed at one instant.
$(\{\text{sun}, \text{rain}\}, 2^E)$	A finite state space with the power set, as in a two-state weather Markov chain. Because E is finite, every subset can be measured and no subtlety arises.
$(\mathbb{N}_0, 2^{\mathbb{N}_0})$	Counting processes: N_t is the number of arrivals by time t.

E need not be numeric. It can be a space of functions, graphs, or measures, which gives function-valued or measure-valued processes. A whole yield curve that evolves over time is a function-valued process. The definition asks nothing of E beyond the existence of $\mathcal{E}$.

§4 Key takeaways

1. The state space must carry a σ-algebra. $\mathcal{E}$ fixes which sets of values a question about X_t can name.
2. Real values default to the Borel σ-algebra. Finite or countable values use the power set.
3. Values need not be numbers. A curve, a graph, or a measure works, provided $\mathcal{E}$ exists.

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5. Ingredient IV: measurability

The condition that makes the probabilities exist

Measurability says that every question about the value of X_t translates into an event:

$$X_t^{-1}(B) = \{\, \omega \in \Omega : X_t(\omega) \in B \,\} \in \mathcal{F} \qquad \text{for all } B \in \mathcal{E}.$$

The left side is the **preimage** of B: the set of outcomes ω for which X_t lands in B. The condition requires that set to be an event.

The condition matters because $\mathbb{P}$ is defined only on $\mathcal{F}$. The expression $\mathbb{P}(X_t \in B)$ is shorthand for $\mathbb{P}\big(X_t^{-1}(B)\big)$. If the preimage is not in $\mathcal{F}$, $\mathbb{P}$ has no value there, and $\mathbb{P}(X_t \in B)$ is undefined. It is not zero, and it is not unknown.

Measurability translates between two kinds of question. A question about values, such as whether the price is below 90, becomes a question about outcomes: which ω produce a price below 90. Measurability guarantees that $\mathbb{P}$ answers the second question.

Measurable at each t versus jointly measurable. The definition requires each X_t to be measurable on its own. That does not make the map $(t, \omega) \mapsto X_t(\omega)$ measurable on $T \times \Omega$. Joint measurability is a strictly stronger property. For an example, take $T = [0, 1]$, let $V \subseteq [0, 1]$ be a set that has no length (Section A), and set $X_t(\omega) = 1$ if $t \in V$ and 0 otherwise. Each X_t is constant in ω, so it is measurable, yet no path can be integrated in t. A path integral such as $\int_0^1 X_t \, dt$ relies on the joint version, which makes every path a measurable function of t and makes the integral a random variable. [Practice] Continuous-time texts usually assume joint measurability rather than derive it. A common route is to assume right-continuous paths, which imply it. Section K defines the σ-algebra on $T \times \Omega$.

§5 Key takeaways

1. $\mathbb{P}(X_t \in B)$ means $\mathbb{P}$ of the preimage $X_t^{-1}(B)$. Measurability guarantees the preimage is an event. Without it, the probability is undefined.
2. Measurability is a precondition for every probability statement about the process, not a technicality.
3. Measurability at each t is weaker than joint measurability in (t, ω). Integrals along a path rely on the joint version, which makes them random variables.

6. Two readings: slices and paths

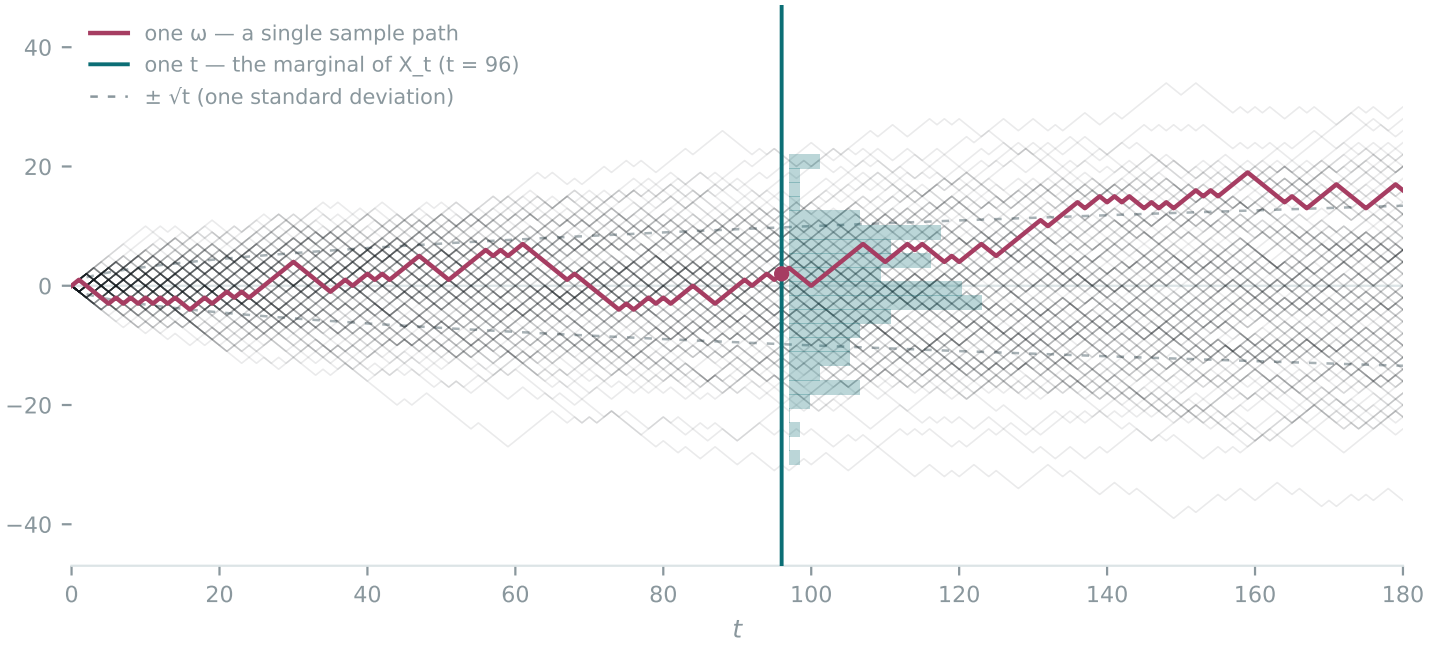
A process $X_t(\omega)$ has two arguments, t and ω. Fixing either one gives a different object, and both views are standard.

- **Fix t, vary ω.** The result is $X_t(\cdot)$, a single random variable. It describes the value at time t across all possible outcomes, and it has a distribution, a mean, and a variance. This chapter calls it the **slice** at t.
- **Fix ω, vary t.** The result is $t \mapsto X_t(\omega)$, a **sample path**, or **path** for short. A path is an ordinary deterministic function of time.

A stochastic process is a random function. Ω indexes the possible paths. $\mathcal{F}$ and $\mathbb{P}$ say how likely each collection of them is. Drawing one ω selects one path, complete for all times at once.

Both readings appear in practice. A Monte Carlo simulation draws many outcomes and generates one path for each. A risk measure at a fixed horizon, such as a 10-day value at risk, uses a slice. A rule that depends on the path the price takes, such as a stop-loss, a drawdown limit, or a barrier option, uses whole paths.

The figure below shows both readings for a **simple random walk**. The walk starts at 0. At each step it moves up or down by 1, with equal probability and independently across steps. Each faint line is one path, produced by one ω from the same Ω. A vertical cut at a fixed t crosses every path. The spread of the crossing points is the distribution of the single random variable X_t, also called the **marginal** distribution at t. That spread has standard deviation $\sqrt{t}$. After 100 steps, the standard deviation is 10 steps, not 100.



§6 Key takeaways

1. Fixing the time gives a random variable, the slice. Fixing the outcome gives a deterministic function of time, the path.
2. A process is a random function: one ω selects one complete path.
3. Risk at a fixed horizon depends on slices. Path-dependent rules and payoffs depend on paths. A model can match every slice and still get the joint behavior across times wrong (§7).

7. Finite-dimensional distributions

The **law** of a process identifies it. The law is the probability measure the process induces on the **path space** E^T. Path space is the set of all functions from T to E, equipped with the product σ-algebra of Section K. Formally, the law is the pushforward of $\mathbb{P}$ through the map $\omega \mapsto (X_t(\omega))_{t \in T}$, which sends each outcome to its path (Section H defines pushforwards). In practice

nobody specifies the law directly. A model instead states the joint distribution of the process at every finite set of distinct times $t_1, \dots, t_n \in T$:

$$\mu_{t_1 \dots t_n}(B) = \mathbb{P}\big((X_{t_1}, \dots, X_{t_n}) \in B\big) \qquad \text{for measurable } B \subseteq E^n.$$

Each $\mu_{t_1 \dots t_n}$ is an ordinary probability distribution on E^n: the joint distribution of n values of the process. The family of all of them, over every finite choice of times, is the process's **finite-dimensional distributions**.

The **Kolmogorov extension theorem** (Kallenberg, 2021) goes in the opposite direction, from distributions to a process. Start from a family of finite-dimensional distributions that is *consistent*, which means two things. First, dropping a time gives the smaller distribution:

$$\mu_{t_1 \dots t_n}(B \times E) = \mu_{t_1 \dots t_{n-1}}(B) \qquad \text{for measurable } B \subseteq E^{n-1}.$$

The left side places no condition on the value at t_n, so it must equal the distribution of the first $n-1$ values. Second, listing the same times in a different order permutes the coordinates of the measure to match. Every process satisfies both conditions automatically, because all of its finite-dimensional distributions come from the same random variables. The theorem says the conditions are also sufficient. It applies to state spaces such as $\mathbb{R}^d$ and, more generally, to any complete separable metric space with its Borel σ-algebra. For such a space, the theorem guarantees a probability space and a process with exactly those finite-dimensional distributions.

The theorem is why Brownian motion can be defined by three properties, without ever writing down Ω. First, $W_0 = 0$. Second, increments over non-overlapping time intervals are independent. Third, $W_t - W_s \sim \mathcal{N}(0, t-s)$ for $s \le t$. These properties fix every finite-dimensional distribution. For times $t_1 < \cdots < t_n$, each W_{t_k} is a sum of independent normal increments, so the vector $(W_{t_1}, \dots, W_{t_n})$ is jointly normal with mean zero and covariance $\mathrm{Cov}(W_s, W_t) = \min(s, t)$. The resulting family is consistent, and the theorem supplies a process.

Distributions at single times are not enough. The finite-dimensional distributions include joint distributions across times, not only the distribution at each single time. To see the difference, let Z be one standard normal draw and set $Y_t = \sqrt{t}\, Z$. At every t, $Y_t \sim \mathcal{N}(0, t)$, the same distribution as W_t. Every path of Y, however, is the same curve rescaled by one draw, so the correlation between Y_1 and Y_4 is 1. For Brownian motion, the covariance formula above gives a correlation between W_1 and W_4 of $1/\sqrt{1 \cdot 4} = 0.5$. The two processes agree at every single time and disagree about every pair of distinct positive times. Derivatives pricing has the same gap. Vanilla option prices across all strikes at each expiry determine the risk-neutral distribution of the asset price at that expiry, one expiry at a time. A barrier or forward-start option depends on the joint distribution across dates, so two models calibrated to the same vanilla prices can still disagree about its price.

The finite-dimensional distributions do not determine path properties. Two processes can share every finite-dimensional distribution while one has continuous paths and the other does not. For an example, let $T = [0, 1]$, let τ be uniform on $[0, 1]$, and compare the zero process with $J_t = \mathbb{1}\{t = \tau\}$, which equals 1 when $t = \tau$ and 0 otherwise. At each fixed t, J_t differs from 0 only when $\tau = t$, which has probability zero. At any n fixed times, the probability that J is non-zero at one of them is at most a sum of n zeros. So the two processes have the same finite-dimensional distributions. Yet every path of the zero process is continuous, and every path of J jumps to 1 at the time τ.

Processes that agree at each fixed t with probability one are called **modifications** of each other. They can still differ in a property that involves uncountably many times at once, such as continuity of the path. A continuous modification therefore needs a separate argument. One example is the Kolmogorov–Chentsov criterion. It produces a continuous modification when the increments $X_t - X_s$ shrink fast enough, on average, as s approaches t (Kallenberg, 2021; Çınlar, 2011). Section K explains why the finite-dimensional distributions cannot see path properties.

§7 Key takeaways

1. A model specifies a process through its finite-dimensional distributions, the joint laws at every finite set of times. The Kolmogorov extension theorem turns a consistent family into a process.
2. Distributions at single times do not determine the finite-dimensional distributions. $\sqrt{t}\, Z$ and Brownian motion agree at every time and differ at every pair of distinct positive times.
3. The finite-dimensional distributions do not determine path properties such as continuity. Path regularity needs its own argument.

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8. Filtrations and adaptedness

The bare definition has no notion of information arriving over time. Almost every application adds one through a filtration.

When the index set T is ordered, a **filtration** is a family of σ-algebras $(\mathcal{F}_t)_{t\in T}$ inside $\mathcal{F}$ that grows with time:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F} \qquad \text{for all } s \leq t.$$

Read $\mathcal{F}_t$ as the events whose truth is settled by time t. The inclusion says that no information is lost over time. An event settled at time s stays settled at every later time t. A probability space together with a filtration, $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in T}, \mathbb{P})$, is a **filtered probability space**. Section B works through a small filtration in full.

A process is **adapted** to the filtration if X_t is $\mathcal{F}_t$-measurable for every t, meaning $X_t^{-1}(B) \in \mathcal{F}_t$ for every $B \in \mathcal{E}$. Every question about the value at time t is then settled by time t, so the value is known by time t. Every process is adapted to its own **natural filtration** $\mathcal{F}_t^X = \sigma(X_s : s \leq t)$, the information revealed by observing X up to time t. Here $\sigma(\cdot)$ denotes the smallest σ-algebra for which every listed random variable is measurable (Section C). Section G makes "known by time t" concrete: an $\mathcal{F}_t^X$-measurable quantity is a function of the observed values X_s for $s \leq t$.

Martingales, stopping times, and stochastic integrals all rely on adaptedness, because each one is a statement about not seeing the future. A **martingale** is an adapted process whose best forecast of any later value, given the information at time s, is its value at s (Section J). A **stopping time** is a random time whose arrival is known when it happens. An example is the first time a price touches a barrier. Formally, the event that the stopping time has occurred by time t belongs to $\mathcal{F}_t$. A **stochastic integral** accumulates the gains of a position that the investor chooses from the information available before each move of the price.

Adaptedness is also the formal version of the rule against lookahead bias in a backtest. A signal at time t must be $\mathcal{F}_t$-measurable, where $\mathcal{F}_t$ is the information actually available at t. A signal computed from anything outside $\mathcal{F}_t$ could not have been known at time t, however good its backtest looks. Three violations are typical. A signal uses a day's closing price to trade at that same close. Fundamentals carry the fiscal period end as their date instead of the release date. A feature is standardized with statistics from the full sample.

§8 Key takeaways

1. A filtration is a growing family of σ-algebras. $\mathcal{F}_t$ holds the events settled by time t.
2. A process is adapted when its value at each time is known at that time. Martingales, stopping times, and stochastic integrals all require it.
3. Lookahead bias is a failure of adaptedness: a signal that is not measurable with respect to the information available at its timestamp.

9. Instantiated

The table applies the definition to five standard processes. Each row names the sample space, the index set, and the state space, then the property that characterizes the process.

Process	Ω	T	$(E, \mathcal{E})$	Character
Simple random walk	$\{-1, +1\}^{\mathbb{N}}$	$\{0, 1, 2, \dots\}$	$(\mathbb{Z}, 2^{\mathbb{Z}})$	X_n sums the first n steps of ω; increments independent
Brownian motion	$C([0, \infty), \mathbb{R})$ with Wiener measure	$[0, \infty)$	$(\mathbb{R}, \mathcal{B}(\mathbb{R}))$	$W_0 = 0$; independent increments, $W_t - W_s \sim \mathcal{N}(0, t - s)$; continuous paths, almost surely nowhere differentiable
Poisson process	Increasing sequences of arrival times	$[0, \infty)$	$(\mathbb{N}_0, 2^{\mathbb{N}_0})$	N_t counts arrivals; independent increments whose distribution depends only on the interval length; right-continuous paths, jumps of size 1
Weather chain	$\{\text{sun}, \text{rain}\}^{\mathbb{N}_0}$	$\{0, 1, 2, \dots\}$	$(\{\text{sun}, \text{rain}\}, 2^E)$	Markov: the next state depends only on the present state
i.i.d. noise	$\mathbb{R}^{\mathbb{Z}}$ with a product measure	$\mathbb{Z}$	$(\mathbb{R}, \mathcal{B}(\mathbb{R}))$	No dependence at all, yet still a stochastic process

In the Brownian row, $C([0, \infty), \mathbb{R})$ is the set of continuous functions from $[0, \infty)$ to $\mathbb{R}$, and Wiener measure is the law of Brownian motion on that set. In the last row, i.i.d. means independent and identically distributed.

The last row shows that the definition requires no dependence between the X_t. It also requires no common distribution and no meaningful order. Each named class of processes adds one such property. Its theorems follow from that addition. Markov chains add a rule for how the next state depends on the present one. Stationary processes add invariance under shifts in time.

Equivalences

Several objects in this chapter have more than one description, and a few pairs that look alike are different. The table collects both kinds.

Objects	Relationship	Section
A family $(X_t)_{t \in T}$; a single map on $T \times \Omega$; a random element of E^T with the product σ-algebra	One process, three descriptions	§1, §7, K
A random vector in $\mathbb{R}^d$; a process with $T = \{1, \dots, d\}$	The same object	§3
A σ-algebra on a finite set; a partition of that set	Same information	B
A σ-algebra; a list of answerable questions; an observer's information	One object, three readings	B

Objects	Relationship	Section
Y is $\sigma(X)$-measurable; $Y = g(X)$ for a measurable g	Equivalent for real-valued Y (Doob–Dynkin)	G
Adapted to the available information; free of lookahead	Equivalent	§8
Law; distribution	Two names for $\mathbb{P} \circ X^{-1}$	H
Probability density function; probability mass function	One Radon–Nikodym density, against Lebesgue measure or against counting measure on a countable set	H
$\mathbb{E}[X \mid \mathcal{G}]$ for square-integrable X; orthogonal projection onto square-integrable $\mathcal{G}$-measurable variables	Equal, up to a null set	J
Measurable at each t; jointly measurable in (t, ω)	Different: joint is strictly stronger	§5
Same distribution at each time; same finite-dimensional distributions	Different: the second is strictly stronger	§7
Same finite-dimensional distributions; same path properties	Different: the first does not imply the second	§7, K

Part I key takeaways

1. A process is a family of functions on Ω, not a sequence of numbers. Numbers appear only after the draw of an outcome ω.
2. The definition draws randomness once, and one ω fixes the whole path. A simulation constructs a path by stepping forward, drawing a shock, and stepping again.
3. The shared probability space turns joint statements across times into events. Random variables on separate spaces have no joint distribution, so dependence between them is undefined.
4. Measurability decides whether $\mathbb{P}(X_t \in B)$ is defined at all.
5. The finite-dimensional distributions identify the law but not the path properties. Information enters through a filtration, and adaptedness is the formal ban on lookahead.

Part II — Appendix: the measure-theoretic vocabulary

Part II defines the terms that Part I uses, starting from bare sets. Sections A–F build the probability space in steps: sets, σ-algebras, measurable spaces, measures, and probability spaces. Sections G–I describe the objects defined on it: measurable functions, distributions, and expectation. Sections J and K cover independence, conditioning, and product spaces. A glossary follows.

Each entry starts with the idea in words, gives the formal definition, and then points back to the place in Part I that relies on it. For proofs, Further reading lists the standard texts in order of difficulty.

Part II uses $(U, \mathcal{U})$ and $(V, \mathcal{V})$ for generic measurable spaces, so that X stays reserved for random variables and processes.

A. Why probability is a theory of sets, not of points

The elementary definition of probability, favorable outcomes divided by total outcomes, needs finitely many equally likely outcomes. Its natural extension gives each outcome a weight and adds the weights, which works whenever the outcomes can be listed one by one. Both fail on a continuum. Drop a point uniformly on $[0, 1]$. Each individual point has probability zero, yet the total probability is one, and no sum of zeros reaches one.

The solution assigns numbers to *sets* of outcomes. The event "the point lands in $[0, \frac{1}{4}]$" has probability $\frac{1}{4}$, with no contradiction. Assigning sizes to sets is the subject of measure theory, which studies consistent notions of size for the subsets of a space.

> Length, area, volume, mass, and probability are all **measures**. They differ only in normalization and interpretation. Probability is a measure with one extra requirement: the whole space has size 1. Every theorem about measures, integrals, and limits therefore applies to probability unchanged.

No consistent rule assigns a size to every subset. Vitali's construction produces subsets of $[0, 1]$ that cannot be given a length. No countably additive, translation-invariant measure that gives $[0, 1]$ length 1 extends to them. Translation-invariant means that shifting a set does not change its size.

The construction works as follows. Call two numbers in $[0, 1)$ equivalent when their difference is rational. The construction uses the axiom of choice to pick one number from each equivalence class. Shifting the chosen set by each rational in $[0, 1)$, wrapping around at 1, gives countably many disjoint copies that together fill $[0, 1)$. Translation invariance forces every copy to have the same length. Countable additivity then makes the total either 0 or infinite, never 1.

The theory keeps countable additivity, because limits depend on it. It gives up measuring every set. It fixes in advance the collection of sets it measures. That collection is a σ-algebra (Section B).

§2 relies on this restriction. In continuous models, $\mathcal{F}$ is smaller than the set of all subsets.

B. σ-algebras: the collection of answerable questions

A σ-algebra is the collection of subsets that a theory measures. The collection must be closed under the operations that combine questions. Combining answerable questions then never produces an unanswerable one.

Definition. Let U be a set. A collection $\mathcal{U}$ of subsets of U is a **σ-algebra** on U if:

(i) $U \in \mathcal{U}$;
(ii) $A \in \mathcal{U} \implies U \setminus A \in \mathcal{U}$ (closed under complement);
(iii) $A_1, A_2, \dots \in \mathcal{U} \implies \bigcup_n A_n \in \mathcal{U}$ (closed under *countable* union).

Members of $\mathcal{U}$ are called **measurable sets**, and in probability they are called **events**. The σ stands for countability.

The three axioms imply more. Axioms (i) and (ii) give $\emptyset \in \mathcal{U}$. De Morgan's laws turn countable unions into countable intersections, so $\mathcal{U}$ is also closed under countable intersection and under set difference. The axioms are minimal. Together they say that $\mathcal{U}$ is closed under every operation built from countably many of its sets.

Reading one: the questions you can ask

Identify each subset $A \subseteq U$ with the yes-or-no question of whether the outcome is in A. The axioms then become rules about questions. If the question for A is admissible, so is the question "not A." If the questions for $A_1, A_2, \ldots$ are admissible, so is the question "at least one of them." A σ-algebra is a collection of questions closed under these logical operations. Any question built from countably many admissible questions is admissible.

The restriction to *countable* unions matters. Suppose the axioms allowed arbitrary unions, and suppose every single point is measurable, as it is in any reasonable model of a continuum. Every subset is the union of its points, so $\mathcal{U}$ would contain every subset. That is the power set, which Section A showed cannot carry length. Countable unions are enough to take limits and few enough to exclude the pathological sets.

Reading two: information

The second reading explains filtrations. Treat $\mathcal{U}$ as the resolving power of an observer. It is the collection of events whose truth the observer can determine. A coarse σ-algebra describes an observer who knows little. A fine one describes an observer who knows more.

- $\{\emptyset, U\}$, the **trivial** σ-algebra, describes an observer who knows only that some outcome occurred. Every real-valued random variable measurable with respect to it is constant.
- 2^U, the **power set**, describes total knowledge. It works for a countable U. On a continuum it is too large to carry length (Section A).
- **The σ-algebra of a partition.** Split U into blocks, and take all unions of blocks. The observer can tell which block occurred, but not which point inside the block. Every finite σ-algebra has this form, so on a finite set a σ-algebra and a partition carry the same information.

Worked example: two coin flips. Let $\Omega = \{\mathrm{HH}, \mathrm{HT}, \mathrm{TH}, \mathrm{TT}\}$. Before any flip, the observer cannot tell any outcomes apart. After the first flip, the observer knows which half contains the outcome. After the second flip, the observer knows the outcome.

```
F0    [ HH  HT  TH  TT ]              knows nothing
F1    [ HH  HT ][ TH  TT ]            knows the first flip
F2    [ HH ][ HT ][ TH ][ TT ]        knows everything
```

The partition gets finer and the σ-algebra grows: $\mathcal{F}_0 \subset \mathcal{F}_1 \subset \mathcal{F}_2$. $\mathcal{F}_0$ contains two events, $\mathcal{F}_1$ contains four, and $\mathcal{F}_2$ contains all 16 subsets of Ω. That increasing chain is a filtration (§8).

Part I uses both readings. The $\mathcal{F}$ of §2 and the $\mathcal{E}$ of §4 are collections of answerable questions, and the filtration of §8 is the information reading.

C. Generation and Borel sets

On a continuum, nobody can list the measurable sets one by one. Instead, one names a few sets that must be measurable and takes the smallest σ-algebra that contains them.

Definition: generated σ-algebra. For any collection $\mathcal{C}$ of subsets of U, $\sigma(\mathcal{C})$ is the smallest σ-algebra that contains $\mathcal{C}$. Equivalently, it is the intersection of all σ-algebras that contain $\mathcal{C}$.

Two facts make the definition work. The power set is a σ-algebra that contains $\mathcal{C}$, so at least one such σ-algebra exists. And any intersection of σ-algebras is again a σ-algebra, because each axiom survives intersection. So the smallest one exists and is unique.

Definition: Borel σ-algebra. $\mathcal{B}(\mathbb{R}) = \sigma(\{\text{open subsets of } \mathbb{R}\})$. The open intervals generate the same σ-algebra, and so do the rays $(-\infty, a]$ with $a \in \mathbb{Q}$. The same construction defines the Borel σ-algebra of any topological space.

The Borel sets are the default answer to the question of which subsets of $\mathbb{R}$ to measure. They include every interval, every open set, every closed set, every countable set, and every set built from these by countable operations. Nearly every set that arises in modeling is Borel. Even so, the Borel sets are a small part of the power set. $\mathcal{B}(\mathbb{R})$ has the cardinality of the continuum, while $2^{\mathbb{R}}$ is strictly larger, so most subsets of $\mathbb{R}$ are not Borel.

> Generation also gives "information" a formal definition. For a random variable X with values in $(E, \mathcal{E})$, the σ-algebra $\sigma(X) = \{X^{-1}(B) : B \in \mathcal{E}\}$ is the information carried by knowing X. The preimages already form a σ-algebra (Section G shows why). It is the smallest σ-algebra on Ω for which X is measurable. For a family of random variables, the preimages of all of them together need not form a σ-algebra. So $\sigma(X_s : s \le t)$ is

the σ-algebra they generate. It is the smallest one for which every X_s with $s \le t$ is measurable. The natural filtration $\mathcal{F}_t^X = \sigma(X_s : s \le t)$ is the information gathered by observing the process up to time t (§8).

The default state space of §4, $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, uses the Borel σ-algebra.

D. Measurable space: structure without numbers

A measurable space is a set together with its collection of measurable subsets. It carries no sizes yet.

Definition. A **measurable space** is a pair $(U, \mathcal{U})$, where U is a set and $\mathcal{U}$ is a σ-algebra on U.

The definition contains no numbers, sizes, or probabilities. A measurable space only declares which questions are legal. It fixes the domain in advance. A measure added later is defined on exactly those sets.

Topology offers a useful parallel. A topological space is a set plus a chosen family of subsets, the open sets, which makes continuity expressible. A measurable space is a set plus a chosen family of subsets, the measurable sets, which makes measurement expressible. The construction is the same. The closure axioms and the purpose differ.

For this reason the definition of a stochastic process requires the state space to be a measurable space $(E, \mathcal{E})$ rather than a bare set E (§4). Without $\mathcal{E}$, the theory defines no set B for which "$X_t \in B$" is a valid question.

E. Measure: a consistent notion of size

A measure assigns a size to each measurable set, and the sizes of disjoint pieces add up to the size of the whole.

Definition. Given a measurable space $(U, \mathcal{U})$, a **measure** is a function $\mu : \mathcal{U} \to [0, \infty]$ with

(i) $\mu(\emptyset) = 0$;
(ii) $\mu(\bigcup_n A_n) = \sum_n \mu(A_n)$ for every sequence of *pairwise disjoint* $A_n \in \mathcal{U}$ (**countable additivity**).

The triple $(U, \mathcal{U}, \mu)$ is a **measure space**.

Countable additivity is the axiom that matters. Finite additivity, under which the sizes of finitely many disjoint pieces add up, is the obvious axiom, but it is too weak. Countable additivity gives **continuity of measure**: if $A_1 \subseteq A_2 \subseteq \cdots$ increase to A, then $\mu(A_n) \uparrow \mu(A)$. The size of a limit is the limit of the sizes. The laws of large numbers, martingale convergence, and the construction of the integral all depend on this property.

Two consequences follow at once. **Monotonicity:** $A \subseteq B$ implies $\mu(A) \le \mu(B)$. **Countable subadditivity:** $\mu(\bigcup_n A_n) \le \sum_n \mu(A_n)$, with no disjointness required. Continuity also holds for decreasing sequences of sets, provided the first set has finite measure. Without that proviso it can fail. The rays $[n, \infty)$ all have infinite length, yet they decrease to the empty set.

The table lists four standard measures.

Measure	Definition	Role
Counting measure	$\mu(A) = \#A$, the number of elements of A	Turns integration into summation
Lebesgue λ	On $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, the unique translation-invariant measure with $\lambda([0,1]) = 1$	Formalizes length, and in $\mathbb{R}^d$ volume
Dirac δ_x	$\delta_x(A) = 1$ if $x \in A$, and 0 otherwise	Puts all the mass at one point
Probability	Any measure with $\mu(U) = 1$	Normalization is its only distinguishing feature

The probability measure $\mathbb{P}$ of §2 is a measure of this kind.

F. Probability space: a measure normalized to one

A probability space is a measure space whose total size is one, with the sizes read as probabilities.

Definition. A **probability space** is a measure space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{P}(\Omega) = 1$. The axioms are $\mathbb{P}(A) \geq 0$ for all $A \in \mathcal{F}$, $\mathbb{P}(\Omega) = 1$, and $\mathbb{P}$ is countably additive over disjoint sequences. In modern form, these are the axioms of Kolmogorov (1933). He stated finite additivity plus a continuity axiom, which together amount to countable additivity.

The specialization adds only new terms:

Symbol	Name	Meaning
Ω	Sample space	The set of all possible complete outcomes
$\omega \in \Omega$	Outcome	One fully specified way the world could turn out
$A \in \mathcal{F}$	Event	A set of outcomes that receives a probability
$\mathbb{P}(A)$	Probability	A number in $[0, 1]$, by monotonicity and normalization

Two terms follow directly and appear constantly. A **null set** is an event A with $\mathbb{P}(A) = 0$. A statement holds **almost surely** (a.s.) if the outcomes where it fails lie inside a null set. In continuous models nearly every theorem is an almost-sure statement, because exceptional outcomes usually exist. The space of continuous paths that carries Brownian motion (§9) contains smooth paths, such as the constant path at zero. The theorem that Brownian paths are nowhere differentiable therefore cannot hold for every outcome. It shows instead that the outcomes where it fails, taken together, form a null set.

A related technicality concerns completeness. A measure space is **complete** if every subset of a null set is measurable, with measure zero. Nothing in the axioms forces completeness, but subsets of null sets are negligible for every practical purpose. [Practice] Authors conventionally complete probability spaces. Continuous-time texts usually also assume that $\mathcal{F}_0$ contains every null set and that the filtration is right-continuous, meaning $\mathcal{F}_t = \bigcap_{u>t} \mathcal{F}_u$. Together these two assumptions are called the **usual conditions**. They ensure, for example, that changing an adapted process on a null set, as passing to a modification does (§7), leaves it adapted.

The build-up, in one table

The table summarizes the steps from a bare set to a probability space.

Object	Name	What it adds
U	A set	No structure. It says which elements exist and nothing else.
$(U, \mathcal{U})$	Measurable space	A σ-algebra: which subsets are legal to ask about. Still no numbers.
$(U, \mathcal{U}, \mu)$	Measure space	Sizes. Each legal subset gets a number in $[0, \infty]$, and the sizes add up consistently.
$(\Omega, \mathcal{F}, \mathbb{P})$, $\mathbb{P}(\Omega) = 1$	Probability space	The same structure, normalized. "Size" now reads as "how likely."

Each step adds one thing and revises none of the earlier structure. Read upward, the table says that probability theory is a special case of measure theory, so every result about measures applies to it unchanged. The probability space is Ingredient I of the definition (§2).

G. Measurable functions, and what a random variable actually is

A measurable function turns every legal question about its output into a legal question about its input. A random variable is such a function on a probability space.

Definition. A function $f : (U, \mathcal{U}) \to (V, \mathcal{V})$ is **measurable** if $f^{-1}(B) \in \mathcal{U}$ for every $B \in \mathcal{V}$. A **random variable** is a measurable function on a probability space.

A common first question is why the condition uses preimages rather than images. Preimages respect every set operation, and images do not. The identities

$$f^{-1}(B^c) = \big(f^{-1}(B)\big)^c, \qquad f^{-1}\Big(\bigcup_n B_n\Big) = \bigcup_n f^{-1}(B_n)$$

hold for every function f. The image version of the first one fails unless f is both one-to-one and onto. If f is constant, for example, the image of any non-empty set is the same single point, so the image of a complement is not the complement of the image. As a result, pulling a σ-algebra back through a function always gives a σ-algebra, while pushing one forward need not. The definition states measurability in the direction that preserves the structure.

The same fact gives a practical test. The collection $\{B \subseteq V : f^{-1}(B) \in \mathcal{U}\}$ is itself a σ-algebra. If it contains a generating family, it contains the whole σ-algebra that family generates. For a real-valued function, the rays generate $\mathcal{B}(\mathbb{R})$ (Section C), so it is enough to check

$$\{f \le a\} = f^{-1}((-\infty, a]) \in \mathcal{U} \qquad \text{for every } a \in \mathbb{R}.$$

Borel measurability then follows. In practice, measurability rarely causes problems. Every continuous function is Borel measurable. Measurable functions stay measurable under sums, products, composition with measurable maps, countable suprema and infima, and pointwise limits. Anything built by ordinary means stays measurable.

> The name *random variable* is historical and misleading. A random variable is a fixed, deterministic function $X : \Omega \to E$. The randomness lies entirely in which outcome ω occurs. The name predates the measure-theoretic foundation. It causes much of the early confusion in the subject.

Doob–Dynkin lemma: information as computability. Section C called $\sigma(X)$ the information in X. The Doob–Dynkin lemma makes that statement exact. Let X be a random variable with values in any measurable space $(E, \mathcal{E})$, and let Y be real-valued. Then Y is $\sigma(X)$-measurable if and only if $Y = g(X)$ for some measurable function $g : E \to \mathbb{R}$ (Kallenberg, 2021). Measurability with respect to the σ-algebra generated by X means exactly "computable from X." To apply the lemma to the natural filtration, take X to be the whole observed path $(X_s)_{s \le t}$. It is a random element of a path space, and the σ-algebra it generates is $\mathcal{F}_t^X$ (Section K). So a real-valued $\mathcal{F}_t^X$-measurable quantity is a function of the observed values X_s for $s \le t$. That is the precise sense of "known at time t" in §8.

Part I treats each X_t as a random variable in this sense (§1, §5).

H. Distributions: the pushforward of $\mathbb{P}$, which drops Ω

A random variable transfers the probability measure from Ω to the state space.

Definition: law, or pushforward. The **law** of X, also called its **distribution**, is the measure $\mu_X = \mathbb{P} \circ X^{-1}$ on $(E, \mathcal{E})$:

$$\mu_X(B) = \mathbb{P}(X^{-1}(B)) = \mathbb{P}(X \in B).$$

The law gives each set of values B the probability of the outcomes that land in it. Measurability of X makes this well defined, and μ_X is again a probability measure.

The law discards everything about Ω. Two random variables on completely different probability spaces can have the same law, and no statement about distributions can tell them apart. For example, one can model a fair coin on $\{0, 1\}$, or on $[0, 1]$ with Lebesgue measure as $\mathbb{1}\{\omega \le \frac{1}{2}\}$. The two models have the same law and serve every purpose that depends only on the law. Models rarely specify Ω explicitly for this reason. §7 applies the same idea to a whole process, whose law is a measure on the path space E^T.

On $\mathbb{R}$, the **cumulative distribution function** $F_X(a) = \mu_X((-\infty, a])$ encodes the law. The rays $(-\infty, a]$ generate $\mathcal{B}(\mathbb{R})$, The intersection of two rays is again a ray. A family closed under intersection in this way is a **π-system**. A uniqueness theorem says that two probability measures that agree on a π-system agree on the whole σ-algebra it generates. So the CDF determines the law.

A density describes a law relative to a reference measure ν on E. Suppose ν is **σ-finite**, meaning E is a countable union of sets of finite ν-measure. Suppose also that μ_X is **absolutely continuous** with respect to ν, written $\mu_X \ll \nu$: every ν-null set is also μ_X-null. The **Radon–Nikodym theorem** then supplies a density f with

$$\mu_X(B) = \int_B f \, d\nu \qquad \text{for every } B \in \mathcal{E}.$$

The density converts the reference measure into the law, one set at a time (Section I defines the integral). With ν equal to Lebesgue measure, f is the probability density function. With ν equal to counting measure on a countable E, f is the probability mass function. The countability matters, because counting measure is σ-finite only on a countable set. The textbook split between continuous and discrete random variables is a choice of reference measure, and the two theories are one theory with different ν.

I. Integration and expectation

The integral averages a function against a measure, and expectation is the integral against a probability measure. The Lebesgue integral has four stages. Each stage extends the previous one to a larger class of functions.

1. **Indicators.** The indicator $\mathbb{1}_A$ of a measurable set A equals 1 on A and 0 elsewhere, and $\int \mathbb{1}_A \, d\mu = \mu(A)$. Integrating an indicator returns the measure of its set.
2. **Simple functions.** A simple function is a finite combination $\varphi = \sum_i c_i \mathbb{1}_{A_i}$ of indicators of measurable sets A_i, with constants $c_i \geq 0$, and its integral is $\sum_i c_i \mu(A_i)$. The value does not depend on how φ is written.
3. **Non-negative measurable f.** $\int f \, d\mu = \sup\{\int \varphi \, d\mu : \varphi \text{ simple}, \, 0 \leq \varphi \leq f\}$. The integral is the best approximation from below by simple functions, and it may be $+\infty$.
4. **General f.** Split $f = f^+ - f^-$ into its positive and negative parts, $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$, and set $\int f \, d\mu = \int f^+ \, d\mu - \int f^- \, d\mu$. Both parts must be finite, which holds exactly when $\int |f| \, d\mu < \infty$.

An integral over a measurable set A restricts the function to that set: $\int_A f \, d\mu = \int f \mathbb{1}_A \, d\mu$. Section H and Section J use this form.

Expectation is this integral with $\mu = \mathbb{P}$:

$$\mathbb{E}[X] = \int_\Omega X(\omega) \, \mathbb{P}(d\omega).$$

The expectation averages X over outcomes, with $\mathbb{P}$ supplying the weights. As Section A requires, the weights attach to sets of outcomes, through the simple functions of stage 2, rather than to single outcomes. The **change-of-variables formula** moves the computation from Ω to the state space, where the law is defined:

$$\int_\Omega g(X) \, d\mathbb{P} = \int_E g \, d\mu_X.$$

It holds for every measurable $g : E \to \mathbb{R}$ that is non-negative or satisfies $\mathbb{E}|g(X)| < \infty$. The formula shows that a density or a mass function alone suffices to compute $\mathbb{E}[g(X)]$, without reference to Ω. When μ_X has density f with respect to ν (Section H), the right side becomes $\int_E gf \, d\nu$. This is the familiar $\int g(x) f(x) \, dx$ for Lebesgue measure, and the sum $\sum_x g(x) f(x)$ for counting measure. A Monte Carlo estimate approximates the left side directly. It averages $g(X(\omega))$ over many simulated outcomes. The mean of a slice in §6 is an expectation in this sense.

> Lebesgue integration replaced Riemann integration because of limits. A pointwise limit of Riemann-integrable functions need not be Riemann integrable. The indicator of the rationals in $[0, 1]$ is the pointwise limit of indicators of finite sets. Each has Riemann integral zero, yet the limit has no Riemann integral. Its Lebesgue integral is zero. Lebesgue's theory provides monotone convergence, for non-negative functions that increase to a limit, and dominated convergence, for functions bounded in absolute value by one integrable function. Each theorem allows exchanging a limit and an integral. Fatou's lemma covers other non-negative sequences with an inequality. The integral of the limit inferior is at most the limit inferior of the integrals. Probability works mostly with limits of sequences of random variables, and these theorems are the main reason its foundation moved to measure theory.

J. Independence and conditioning

Independence says that learning one event does not change the probability of another. Conditioning says how to update an estimate when partial information arrives.

Events A and B are **independent** when $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Two sub-$\sigma$-algebras $\mathcal{G}, \mathcal{H} \subseteq \mathcal{F}$ are independent when this holds for every $A \in \mathcal{G}$ and $B \in \mathcal{H}$. Random variables are independent when the σ-algebras they generate are independent.

Independence is a property of the measure, not of the sets. The same two events can be independent under one $\mathbb{P}$ and dependent under another. "The first flip is heads" and "the second flip is heads" are independent for fair, independent coins. They are dependent under a measure in which the second flip tends to repeat the first.

Conditioning uses the information reading of σ-algebras (Section B). Conditional expectation given a single event A with $\mathbb{P}(A) > 0$ is elementary. It is $\mathbb{E}[X \mid A] = \mathbb{E}[X\mathbb{1}_A]/\mathbb{P}(A)$, the average of X over A. The theory uses conditional expectation given a σ-algebra, which applies that idea to every event the σ-algebra resolves at once.

Definition: conditional expectation. For an integrable X and a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$, $\mathbb{E}[X \mid \mathcal{G}]$ is *any* random variable Z that is

(i) $\mathcal{G}$-measurable, and
(ii) satisfies $\int_G Z \, d\mathbb{P} = \int_G X \, d\mathbb{P}$ for every $G \in \mathcal{G}$.

Such a Z exists and is unique up to a null set. The Radon–Nikodym theorem (Section H) supplies it. For non-negative X, the map $G \mapsto \int_G X \, d\mathbb{P}$ is a measure on $\mathcal{G}$ that is absolutely continuous with respect to $\mathbb{P}$ restricted to $\mathcal{G}$. Its density is $\mathcal{G}$-measurable and satisfies clause (ii), so it is Z. The same argument applies to the positive and negative parts of a general X.

The two clauses have plain meanings. Clause (i) says the answer may use only the information in $\mathcal{G}$. Clause (ii) says that on every event $\mathcal{G}$ can resolve, the answer has the same integral as X, so it matches the averages at the available resolution. Together they make $\mathbb{E}[X \mid \mathcal{G}]$ the best estimate of X that uses only the information in $\mathcal{G}$. For a square-integrable X, meaning $\mathbb{E}[X^2] < \infty$, "best" has an exact meaning. Take the inner product of two square-integrable random variables to be the expectation of their product. $\mathbb{E}[X \mid \mathcal{G}]$ is then the orthogonal projection of X onto the square-integrable $\mathcal{G}$-measurable random variables, so it has the smallest mean squared error among them.

In the two-coin example of Section B, let the flips be fair and independent, so each outcome has probability $\frac{1}{4}$. Let X be the number of heads and take $\mathcal{G} = \mathcal{F}_1$. On the block $\{\mathsf{HH}, \mathsf{HT}\}$, X equals 2 or 1 with equal probability, so its average there is 1.5. On $\{\mathsf{TH}, \mathsf{TT}\}$ the average is 0.5. So $\mathbb{E}[X \mid \mathcal{F}_1]$ equals 1.5 on the outcomes where the first flip is heads and 0.5 where it is tails.

$\mathbb{E}[X \mid \mathcal{G}]$ is a random variable, not a number. It still depends on ω, at the coarser resolution of $\mathcal{G}$. In the example, it takes one value on each block of the partition. That property lets the **martingale** condition describe a process. An adapted, integrable process is a martingale when

$$\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s \qquad \text{for all } s \le t.$$

Given everything known at time s, the best forecast of the value at time t is the value at time s. §8 names martingales among the objects that rely on adaptedness.

K. Product spaces

A product space combines two measurable spaces. Given $(U, \mathcal{U}, \mu)$ and $(V, \mathcal{V}, \nu)$, the **product σ-algebra** $\mathcal{U} \otimes \mathcal{V}$ is the σ-algebra on $U \times V$ generated by the rectangles $A \times B$ with $A \in \mathcal{U}$ and $B \in \mathcal{V}$. When μ and ν are σ-finite, exactly one **product measure** $\mu \otimes \nu$ satisfies $(\mu \otimes \nu)(A \times B) = \mu(A)\nu(B)$. The **Fubini–Tonelli theorem** then permits swapping the order of a double integral: always for non-negative integrands, and for general integrands once the absolute value is integrable.

Part I relied on this construction in two places.

- **Joint measurability (§5).** To call $(t, \omega) \mapsto X_t(\omega)$ measurable, $T \times \Omega$ needs a σ-algebra. The standard choice is the product of the Borel σ-algebra on T with $\mathcal{F}$. Joint measurability lets Fubini–Tonelli exchange a time integral with an expectation, as in $\mathbb{E}\big[\int_0^1 X_t \, dt\big] = \int_0^1 \mathbb{E}[X_t] \, dt$, for non-negative X or when $\mathbb{E}\big[\int_0^1 |X_t| \, dt\big] < \infty$. Occupation-time arguments, which measure how long a path spends in a set of values, need it. Stochastic integration needs it too, in a stronger form that also respects the filtration.
- **Path space (§7).** The law of the whole process is defined on E^T with the product σ-algebra. The **cylinder sets** generate this σ-algebra. A cylinder set is a set of paths that constrains the value at finitely many indices only, such as $\{x \in E^T : x(t_1) \in B_1, \dots, x(t_n) \in B_n\}$. Each cylinder set is a finite intersection of sets of the form $\{x : x(t) \in B\}$. So the product σ-algebra is also the smallest one for which every coordinate map $x \mapsto x(t)$ is measurable. Two consequences follow. The map $\omega \mapsto (X_t(\omega))_{t \in T}$ is measurable exactly when every X_t is, so a process is the same thing as a random element of E^T. And the σ-algebra that random element generates is $\sigma(X_t : t \in T)$.

  This construction explains both results of §7. The finite-dimensional distributions are exactly the probabilities of cylinder sets, and the cylinder sets generate the product σ-algebra. The intersection of two cylinder sets is again a cylinder set, so the cylinder sets form a π-system. By the uniqueness theorem of Section H, the finite-dimensional distributions therefore determine the law.

  The finite-dimensional distributions cannot determine path properties, for a related reason. Every event in the product σ-algebra depends on the path at only countably many indices. The sets with that property form a σ-algebra, because a countable union of countable index sets is countable. That σ-algebra contains every cylinder set. So it contains the whole product σ-algebra. When T is an interval, no countable set of indices can decide continuity. Any countable set of values is consistent with both a continuous path and a discontinuous one. So the set of continuous paths is not in the product σ-algebra. “The process has continuous paths” is therefore not an event there. Continuity has to come from choosing a good modification, not from computing a probability.

Glossary

Symbol	Name	Read it as
Ω	Sample space	The set of complete outcomes. One ω is one whole history
$\mathcal{F}$	σ-algebra	The events that receive a probability. Equivalently, the information available
$\mathbb{P}$	Probability measure	A countably additive size on $\mathcal{F}$, normalized so that $\mathbb{P}(\Omega) = 1$
$(U, \mathcal{U})$	Measurable space	A set plus its legal subsets, with no numbers yet
$(U, \mathcal{U}, \mu)$	Measure space	A measurable space plus sizes in $[0, \infty]$
$\mathcal{B}(\mathbb{R})$	Borel σ-algebra	Generated by the open sets. These are the default measurable subsets of $\mathbb{R}$
$\sigma(\mathcal{C})$	Generated σ-algebra	The smallest σ-algebra containing $\mathcal{C}$
$\sigma(X)$	σ-algebra of X	$\{X^{-1}(B) : B \in \mathcal{E}\}$. These are the events settled by observing X
$X : \Omega \to E$	Random variable	A measurable function. It is deterministic, and ω is the only thing that varies
$\mu_X = \mathbb{P} \circ X^{-1}$	Law, or distribution	$\mathbb{P}$ transferred to the state space, with Ω dropped
$\mathbb{E}[X] = \int X \, d\mathbb{P}$	Expectation	The average over outcomes, weighted by probability
$\mathbb{E}[X \mid \mathcal{G}]$	Conditional expectation	The best estimate of X using only what $\mathcal{G}$ resolves. It is itself random
$(\mathcal{F}_t)_{t \in T}$	Filtration	Information that grows with time, with $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$
a.s.	Almost surely	True outside a set of probability zero
$\mu \ll \nu$	Absolute continuity	Every ν-null set is μ-null. For σ-finite ν, this is the condition for a density to exist

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Further reading

All of the material in this chapter is standard. The texts below supply the proofs, ordered roughly by how much measure theory they assume.

The measure-theoretic foundation.

- **Williams, D. (1991).** *Probability with Martingales.* Cambridge University Press. — The gentlest rigorous introduction. Read it if Part II was too fast. It covers σ-algebras, measure, integration, and conditional expectation in about a hundred pages. The rest covers discrete-time martingales.
- **Billingsley, P. (2012).** *Probability and Measure*, Anniversary ed. Wiley. [[paywalled]] — The standard graduate reference for the measure theory itself. It is slower and more complete than Williams. Use it to find the full construction of a result, not only its statement.
- **Durrett, R. (2019).** *Probability: Theory and Examples*, 5th ed. Cambridge University Press. — Covers the same ground, then continues into Brownian motion and martingales. The author posts the full text free. That makes it the most practical of the three to consult while reading.

Processes proper.

- **Çınlar, E. (2011).** *Probability and Stochastics.* Springer, Graduate Texts in Mathematics 261. [[paywalled]] — It treats processes from the first page and does not reach them only after a measure-theory course. It is the closest match to how this chapter presents the definition.

- **Kallenberg, O. (2021).** *Foundations of Modern Probability*, 3rd ed. Springer. [[paywalled]] — The comprehensive reference. It is too terse for a first reading. It states the Kolmogorov extension theorem (§7), the Kolmogorov–Chentsov continuity criterion, and the Doob–Dynkin lemma (Section G) in their general forms.
- **Øksendal, B. (2003).** *Stochastic Differential Equations: An Introduction with Applications*, 6th ed. Springer. [[paywalled]] — The next text to read once the objects here are clear. It covers the Itô calculus, which builds on them. It is deliberately light on measure-theoretic detail, which suits a reader who has finished Part II.

The historical source.

- **Kolmogorov, A. N. (1933).** *Grundbegriffe der Wahrscheinlichkeitsrechnung* (trans. *Foundations of the Theory of Probability*, Chelsea, 1956). Springer. — The monograph that made probability a branch of measure theory. It is the source of the axioms in Section F. It is short and more readable than its reputation suggests.

If you only read these. Recommendation: Williams (1991) for the foundation, then Çınlar (2011) for processes. Durrett (2019) is the free alternative to both.

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